International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2018-0004
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Human Kinetics
ORIGINAL INVESTIGATION

The Influence of Sport-Field Properties on Muscle-Recruitment Patterns and Metabolic Response

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Purpose: To investigate different sport-field properties' influence on muscle-recruitment patterns and metabolic response during a series of running and agility drills. ***Methods:*** Eleven male athletes were fitted with a standard multipurpose training shoe. The test protocol consisting of 4 high-intensity trials with 60-s rests between trials performed on 2 fields with different properties. Time-dependent field properties were measured using the American Standards for Testing and Materials protocol (F-1936). A 30-m pretest and posttest sprint determined fatigue and player performance. Electromyography (EMG) recorded muscle activity for vastus medialis, biceps femoris, gastrocnemius medial head, and tibialis anterior, and metabolic activity analyzed maximal oxygen consumption, heart rate, respiratory exchange ratio, metabolic equivalent, and energy expenditure. ***Results:*** A difference was calculated for muscle activity across trials ($P = .01$) for both surfaces. Muscle activity was <13% on the field with less energy return ($P = .01$). Metabolic components (maximal oxygen consumption, heart rate, respiratory exchange ratio, metabolic equivalent, and energy expenditure) were significantly different across trials ($P = .01$) but not significantly different between fields. The participants completed the agility course (5.2%) faster on the field with greater energy return, while caloric expenditure was similar between fields. ***Conclusions:*** The findings indicate that field mechanical properties influence muscle-activation patterns. The field demonstrating the greatest magnitude of energy return produces the lowest sprint and agility course times; however, performing on a field exhibiting unfamiliar mechanical properties could cause the athlete to produce atypical movement patterns that might contribute to overuse of the neuromuscular system.

Keywords: neuromuscular, electromyography, metabolism, fatigue, energy return

Sport field time-dependent properties, which influence player movement strategies,[1] can differ significantly even between similar field types. Identifying the effect of these factors on the human body could provide invaluable information for developing athlete preparation and training protocols to address the incidence of overuse injury and performance enhancement. The influence of these mechanical properties can vary from field-to-field depending on several factors: field composition (soil, rubber, sand, grass, or synthetics), surface structural elements, base preparation, installation process, underlayment type, surface temperature, and surface condition (maintained or not maintained to standard), to name a few. The ability of a surface to absorb impact force influences the magnitude of the loading rate on the musculoskeletal system. A sport field with high absorption qualities could be beneficial for reducing impact force but performance would most likely suffer. Conversely, a surface with low impact attenuation qualities is defined as a "stiff" surface. It is conceivable to think that a stiffer surface that produces a large mechanical energy return component would equate to a large performance gain. This is not necessarily the case because energy transfer occurring outside an optimal range would have an adverse effect on performance due to the viscoelastic property of human tissue.[2,3] Ultimately, a surface with less energy return requires a greater amount of work by the athlete to maintain a high output of performance.[4] The objective of the study examines human performance factors while athletes perform a high-intensity activity on athletic fields with different mechanical properties.

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Methods

Participants

A total of 11 male athletes signed a consent form before participating in the Kennesaw State University Internal Review Board–approved research study. The sport classification of the volunteers was 4 lacrosse, 3 soccer, 2 rugby, and 2 football players. Participants' physical characteristics (mean [SD] age = 20.7 [1.1] y, weight = 86.6 [4.9] kg, and height = 1.8 [0.1] m) were recorded. The test protocol was designed to induce muscle fatigue through physical activity so a high level of fitness was mandatory. In addition to meeting a stringent qualification criterion, the participants completed a physical fitness questionnaire to ensure their safety and to determine their familiarity level with a variety of agility and running drills. The participants were provided with nutritional instruction 48 hours prior to testing to ensure hydration and glycogen levels were at optimal levels. Each athlete was instructed to fast 2 hours prior to testing and not to engage in exercise, sport practice, or any other strenuous activity 48 hours prior to testing.

Study Design

The study follows a randomized, repeated-measures experimental design. Two outdoor athletic fields with different time-dependent properties were selected to assess muscle activity: vastus medialis (VM), biceps femoris (BF), gastrocnemius medial head, and tibialis

anterior (TA); and cardiopulmonary factors: maximal oxygen consumption (VO_2max), heart rate (HR), respiratory exchange ratio (RER), metabolic equivalent of task (MET), and energy expenditure. The test course (20×20 m^2 area) consisted of 8 different agility skills that required the participants to incorporate acceleration, change of directions, and resistance-incorporated running drills. Figure 1 shows the detailed layout of the course.

Procedures

Performance Testing Protocol. On test days, the participants were fitted with a multipurpose training shoe (Men's Ultimate Turf Trainer; Under Armour, Baltimore, MD). This was followed by the lead-practitioner attaching surface electromyography (EMG) electrodes to the predetermined sites of the lower extremity. Afterward, the participant ran through the test course at a low-intensity pace for 5 minutes, minus the farmer's carry and weighted sled pull. This accomplished 2 objectives: (1) familiarization with the agility course and (2) a warm-up session. After the preparatory phase, the athlete performed the maximum voluntary contraction (MVC) protocol for each muscle selected for testing. Next, the gas analyzer and face mask were fitted to the athlete and this was followed by a 5-minute acclimation period prior to testing to confirm correct fit of the devices and to ensure the participant was comfortable wearing the devices while performing the test protocol. The testing session consisted of 4 consecutive trials (T1, T2, T3, and T4) with a 60-second rest between each trial. Lastly, the participants were instructed to complete the course at maximum effort as quickly as possible. The athletes performed the test protocol on each athletic field 7 days apart to allow appropriate recovery time and minimize a carryover effect. Additionally, the participants performed each test protocol at the same time of day on the different fields to minimize the influence of temperature and humidity on performance and metabolic response.

Instrumentation. Six digital (120 frames/s @ 1080 p) cameras (GoPro Hero4, San Mateo, CA) were positioned to record the 30-m sprint and agility course. The cameras were synchronized using a GoPro Wi-Fi remote. Sprint times and agility course times were calculated using video editing software (Adobe Premiere Pro, San Jose, CA) for accuracy since we initiated and completed all data collection events when the first body part crosses the start and finish lines.

Surface EMG data were collected with a direct transmission system (1000 Hz) incorporated with myoResearch software (Noraxon USA, Inc, Scottsdale, AZ). The raw dynamic EMG, and MVCs for each muscle, were filtered using a fourth-order Butterworth bandpass (high pass with 20 Hz cutoff and low pass with 500 Hz cutoff) processed with 20 ms root mean square smoothing window algorithms. The EMG signals were recorded with bipolar surface electrodes (Noraxon, USA, Inc) for the VM, BF, gastrocnemius medial head, and TA. Electrode placement for each participant was performed by the same practitioner throughout the study. The raw EMG data were exported to MATLAB R2017a (MathWorks, Inc, Natick, MA) for signal analysis and processing.

A portable gas exchange system (K4b2; COSMED Inc, Concord, CA) was worn by the participants during the test protocol to

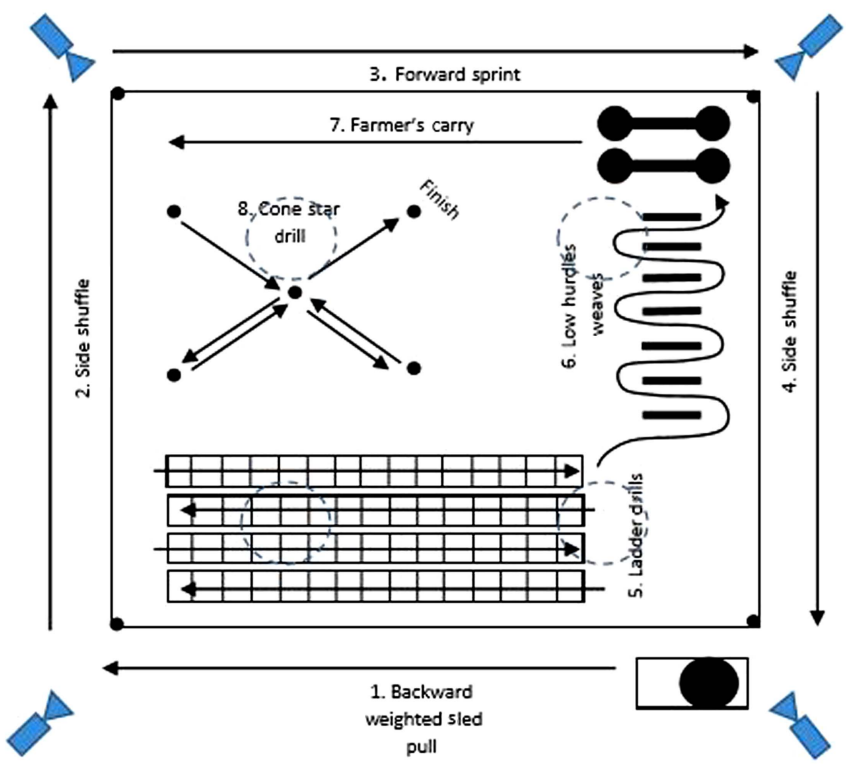

Figure 1 — Agility circuit layout. *Note*. The activity began with a 20-m weighted (40 kg) sled pull, followed by a shuffle, sprint, and shuffle for 20 m each. The 4 corners was followed by 2-foot run and side shuffle ladder drills. The athlete proceeded to a 10-m weave drill utilizing low hurdles and then carried a pair of 10-kg farmer's-carry equipment for 15 m. The last drill was a star drill, which entailed forward and backward sprints. The outer cones for the star drill were positioned 8 m apart. The dashed circles identify the location of the field test sites using the Advanced Artificial Athlete device. The 4 cameras were positioned at each corner, with camera 1 positioned to view both start and finish lines. The sprint course incorporated 2 additional cameras positioned at the start and finish lines.

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collect volume of oxygen (VO_2) and volume of carbon dioxide (VCO_2) samples. The device utilizes a breath-by-breath measurement of gas exchange through a rubberized facemask and and the collected data were exported to OMNIA v 1.4 (COSMED Inc.) for analysis, interpretation, and reporting.

A sport field testing protocol stipulated by the American Standards for Testing and Materials (ASTM) was used to measure mechanical properties of the athletic fields. The Advanced Artificial Athlete recorded force reduction (in percentage), vertical deformation (in millimeters), and energy restitution (ER; in percentage) (Labosport France, Le Mans, France). The field test procedure (ASTM-1936) was performed in each quadrant of the test areas prior to data collection (Figure 1).

Baseline Measurement Protocols. For muscle activation, an MVC protocol established baseline measures of the selected lower extremity muscles. EMG values were expressed as a percentage of amplitude from the MVC. EMG reference values were calculated for each muscle using the average of the 3 peak EMG signals which represented 100% MVC. For cardiopulmonary activity, VO_2 and VCO_2 samples were collected during a graded treadmill protocol (2.5% grade increase every 2 min), also a 300-m shuttle-run test protocol to establish anaerobic capacity baseline measures. VO_2-max, HR, RER, MET, and EE reference values were calculated and used to compare values obtained during the test sessions.

Statistical Analysis

Group means and SDs were used to determine physiological differences influenced by sport field time-dependent properties. An alpha level of .05 (n = 11) was adopted throughout the analysis to minimize any type I statistical errors. For muscle recruitment, a 3-way repeated-measures analysis of variance (ANOVA) examined muscle activity (VM, BF, MG, and TA) differences while an athlete performed 4 agility course circuits on 2 separate sports fields. The sample design consisted of 1 between-subject independent variable (field type) and 2 within-subject independent variables (muscle group × agility trial). The observed power for the analysis was .688 for field type effect, .773 for muscle group effect, .997 for agility trial effect, .999 for muscle group × agility trial interaction, and .892 for field type × agility trial interaction. Between-subject parameter estimates compared the treatment (field type) effect on muscle activation patterns. For metabolic response, 2-way repeated-measures ANOVA examined differences between field type and agility trials for the selected dependent variables (VO_2, HR, MET, RER, and energy expenditure). For the sample design, within-subjects effects (trial), between-subjects effects (field), and between-subjects interaction effects (field type × trial) were examined. Follow-up pairwise comparisons were performed, where appropriate. A Pearson correlation analysis examined muscle fatigue patterns where the fields represented independent variables and muscle frequency represented the dependent variable. Data were analyzed with the statistic software SPSS v 24.0 (IBM Corp, Armonk, NY).

Results

The test fields are referenced as either *Field A* or *Field B.* Ambient temperature for *Field A* data collection was 22.3°C (3.1°C) and 23.6°C (2.5°C) for *Field B.* Relative humidity for *Field A* testing was 58.3% (3.3%) and 61.4% (3.9%) for *Field B* during data collection. No significant difference was determined for ambient temperature or relative humidity.

Mechanical Testing

Mean ER for *Field A* was lower (28.23% [2.14]%) than the ER for *Field B* (41.63% [1.98%]), $t(3) = 11.14$, $P < .011$, $d = 6.49$. The calculated t test, $t(3) = 3.82$, $P = .023$, $d = 3.06$, for force reduction on *Field A* (50.53% [1.78%]) and *Field B* (54.91% [1.08%]) show a significant difference. Mean vertical deformation was also less for *Field A* (5.68 [0.42] mm) compared with *Field B* (7.13 [0.26] mm), $t(3) = 5.75$, $P = .02$, $d = 4.23$. International field performance standards are presented in Table 1.

Muscle-Recruitment Patterns

The 3-way interaction was not significant between the fields, muscle groups, and evaluation trials. Follow-up statistical analyses identified signficant 2-way interactions and main effects. Initially, we analyzed muscle activity by combining the 4 muscle group recordings for *Field A* and *Field B*. The means and SDs are summarized in Table 2.

A 2-way repeated-measures ANOVA revealed an interaction effect between muscle group and agility trial, $F(9, 180) = 6.05$, $P < .001$, $\eta_p^2 = .232$, for mean muscle activity from the group of muscles under investigation. A follow-up simple main effect analysis indicated a significant difference between fields, $F(1, 20) = 5.78$, $P = .026$, $\eta_p^2 = .224$. A 2-way repeated-measures ANOVA revealed an interaction effect between the fields and evaluation trials, $F(3, 60) = 4.72$, $P = .004$, $\eta_p^2 = .197$, for mean muscle activity from the group of muscles under investigation. A follow-up simple main effect analysis indicated a significant difference between muscle groups, $F(3, 60) = 7.06$, $P < .001$, $\eta_p^2 = .260$. Further analysis revealed significant muscle activity mean difference between fields (mean = 0.65, SD = 0.58) for the BF muscle group, $F(1, 20) = 4.51$, $P = .039$, $\eta_p^2 = .122$; and significant difference for the TA muscle group, $F(1, 20) = 3.33$, $P = .031$,

Table 1 International Sport-Field Performance Standards

	Field standards		
	IRB[5]	FIFA[6] (1 star)	FIFA (consistency)
FR, %	55–70	55–70	<10%
VD, mm	5.5–10	4–9	<15%
ER, %	20–50	—	—

Abbreviations: ER, energy restitution; FIFA, Fédération Internationale de Football Association; FR, force reduction; IRB, International Rugby Board; VD, vertical deformation. Note: FIFA consistency describes the acceptable range between test sites on the field.

Table 2 Comparison of Means (SD) for Overall Muscle Activity During the 4 Trials on Different Athletic-Field Surfaces

	Agility trial			
	T1	T2	T3	T4
Field A	740.82 (144.21)	611.14 (116.14)	499.91 (99.21)	390.91 (88.39)
Field B	566.63 (105.39)	447.81 (86.04)	381.27 (85.37)	321.45 (70.31)

Table 3 Comparison of Means (SD) for Individual Muscles Between the Sport-Field Surfaces

	Agility trial			
Muscle	T1	T2	T3	T4
Field A				
VM	236.91 (48.04)	180.73 (52.59)	141.64 (48.77)	101.46 (32.49)
BF	245.91 (51.38)	201.09 (50.07)	166.09 (43.43)	119.55 (42.65)
MG	147.91 (38.31)	118.55 (34.05)	105.64 (34.00)	92.465 (33.10)
TA	140.09 (37.27)	110.64 (33.75)	86.55 (32.56)	77.46 (29.94)
Field B				
VM	161.09 (36.64)	122.73 (12.52)	103.64 (14.38)	82.28 (18.74)
BF	167.91 (32.02)	128.64 (32.51)	107.00 (26.99)	84.09 (12.47)
MG	124.09 (32.06)	99.91 (22.95)	86.46 (24.78)	78.55 (26.06)
TA	113.55 (16.44)	96.55 (16.93)	84.18 (16.42)	76.55 (15.12)

Abbreviations: BF, biceps femoris; MG, gastrocnemius medial head; TA, tibialis anterior; VM, vastus medialis.

Table 4 Comparison of Normalized EMG (%MVC) Means (SD) for the Selected Muscles

	Agility trial			
Muscle	T1	T2	T3	T4
Field A				
VM	43.21 (15.42)	32.65 (11.11)	25.76 (8.18)	21.52 (5.92)
BF	49.34 (17.83)	40.17 (14.82)	33.21 (10.55)	24.39 (9.15)
MG	36.67 (4.13)	29.23 (2.98)	26.45 (2.84)	22.27 (2.43)
TA	34.34 (19.49)	26.71 (12.32)	19.17 (6.33)	18.53 (5.96)
Field B				
VM	32.82 (8.25)	26.44 (7.57)	22.62 (5.61)	17.23 (2.93)
BF	33.11 (5.79)	26.49 (4.35)	21.74 (3.56)	16.82 (3.35)
MG	31.78 (6.14)	25.83 (3.62)	21.26 (3.91)	19.17 (3.43)
TA	27.65 (11.92)	23.27 (10.11)	20.32 (9.82)	19.78 (7.99)

Abbreviations: BF, biceps femoris; EMG, electromyography; MG, gastrocnemius medial head; MVC, maximum voluntary contraction; TA, tibialis anterior; VM, vastus medialis.

$\eta_p^2 = .257$. Table 3 summarizes mean individual muscle activity during the agility test protocol across trials on the selected fields.

An MVC protocol determined a baseline for each muscle selected for analysis. Table 4 presents means and SDs percentages of muscle activity relative to the maximum recorded values for the duration of the agility circuit.

Sprint and Circuit Times

A repeated-measures ANOVA determined an interaction effect between field type and sprint test, $F(1, 10) = 7.08$, $P = .02$, $\eta_p^2 = .346$. The pretest and posttest times were analyzed independently to identify any differences associated with the time-dependent field properties (Table 5). The mean pretest 30-m sprint time was slower on *Field A* than *Field B*, $t(10) = 19.634$, $P = .01$, $d = 1.44$. The calculated paired t test, $t(10) = 13.23$, $P = .01$, $d = 1.76$, also revealed the 30-m mean posttest time on *Field A* was slower than the sprint time on *Field B*. The agility test protocol was divided into 4 trials (circuits) for each field. The mean time to complete 1 agility circuit on *Field A* was greater than *Field B*, $t(10) = 4.53$, $P = .02$, $d = 1.29$.

Cardiopulmonary and Metabolic Response

The cardiopulmonary components were analyzed for the individual trials (Table 6). A 2-way repeated-measures ANOVA determined no interaction effect for VO_2max, HR, RER, or MET. Across trials, average RER for both *Field A* and *Field B* demonstrated a main effect difference, $F(3, 60) = 32.86$, $P = .01$, $\eta_p^2 = .641$, however, no significant difference between fields. Pairwise comparisons for RER determined a significant difference ($P < .05$) exists between T2 and the other trials. Similarly for VO_2max, no interaction effect was determined but a difference for (trial) main effect, $F(3, 60) = 3.45$, $P = .02$, $\eta_p^2 = .147$, was calculated, while no significant difference was determined between fields. Further alaysis of pairwise comparisons indicate greater oxygen consumption for T2 on *Field A* ($P = .02$). A main effect was calculated for MET, $F(3, 60) = 2.63$, $P = .03$, $\eta_p^2 = .163$, with no interaction effect across trials. Again, pairwise comparisons revealed MET was greater during T2 compared with T1 and T4 ($P = .02$). For comparison, baseline measures for peak VO_2max (48.14 [4.55]) and HR (189.57 [6.38]) values were calculated during the anaerobic capacity test protocol.

Table 5 Comparison of Means (SD) for Pretest and Posttest 30-m Sprint Times (s) and the 4 Agility-Circuit Times

	Pretest	1	2	3	4	Posttest
Field A	4.665 (0.26)	91.12 (3.74)	92.09 (4.09)	93.78 (4.99)	95.72 (5.25)	5.181 (0.31)
Field B	4.307 (0.24)	87.25 (3.05)	88.22 (2.88)	89.87 (2.96)	89.88 (3.21)	4.734 (0.22)

The portable gas analyzer measured VO_2 and VCO_2 exchange samples and calculated energy expenditure by indirect calorimetry. An interaction effect was not determined by the 2-way repeated-measures ANOVA for energy expenditure across trials and between field types. However, Figure 2 illustrates a simple main effect between *Field A* (102.53 [9.21]) and *Field B* (104.41 [11.45]) for T2, $t(10) = 2.42$, $P = .03$, $r = .92$.

Muscle Fatigue Effect

Median frequency in the time domain was used to identify muscle fatigue patterns across field types. First, data were normalized in the time domain prior to analysis. A Pearson product–moment

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Table 6 Comparison of Means (SD) for Cardiopulmonary and Metabolic Components

Metabolic component	Agility trial T1	T2	T3	T4
Field A				
VO_2	37.59 (3.49)	38.03 (3.66)	37.93 (4.46)	37.21 (4.62)
HR	176.61 (6.71)	180.14 (6.62)	183.71 (7.05)	182.71 (8.06)
RER	1.20 (0.11)	1.12 (0.09)	1.05 (0.08)	1.03 (0.08)
MET	10.74 (1.04)	10.94 (1.06)	10.83 (1.26)	10.65 (1.31)
EE	25.91 (2.48)	25.25 (4.35)	25.73 (4.21)	25.62 (4.44)
Field B				
VO_2	38.95 (4.14)	40.47 (3.46)	39.13 (3.51)	38.34 (4.62)
HR	173.45 (8.66)	178.88 (6.65)	179.88 (6.68)	181.22 (8.62)
RER	1.16 (0.08)	1.09 (0.08)	1.02 (0.06)	0.99 (0.04)
MET	11.12 (1.08)	11.61 (0.88)	11.45 (1.01)	10.95 (1.23)
EE	24.83 (4.85)	27.57 (4.92)	26.42 (4.94)	25.32 (4.69)

Abbreviations: EE, energy expenditure; HR, heart rate; MET, metabolic equivalent of task; RER, respiratory exchange ratio.

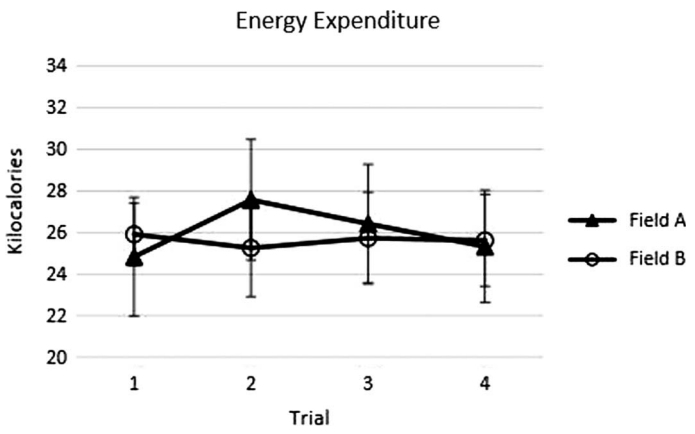

Figure 2 — Mean energy expenditure compared between sport fields for each trial.

correlation was computed to assess the relationship between mean frequency and field types. There were weak correlations calculated between both VM activity (r = .298, N = 87, P = .01) and gastrocnemius medial head activity (r = .284, N = 87, P = .01). There was also a weak correlation between field type and BF activity (r = .332, N = 87, P = .01). A moderate correlation was calculated between field type and the TA muscle activity (r = .421, N = 87, P = .01). A series of scatterplots summarize the results (Figure 3).

Discussion

It has been shown that human movement strategies are influenced by surface mechanical properties.[1] Our research provides additional support to this concept by analyzing neuromuscular and metabolic response on sports fields exhibiting different mechanical properties. The findings suggest muscle activation and recruitment patterns are influenced by the time-dependent properties of a surface which causes the body to adapt a specific set of movement parameters. Similar to previous research, we used lower extremity muscle activation patterns and metabolic activity to evaluate the player–surface interaction.[4,7] In a sport performance scenario where a greater portion of energy from the athlete to the surface is returned in the form of mechanical energy, the athlete could conceivably perform a projected workload with less effort.[7,8] Regardless, the mechanical energy returned to the athlete has to be dealt with by the musculoskeletal system,[9] and the consequence is either an improvement or deterioration in performance. For example, a result of the musculoskeletal system mismanaging the returned energy could be premature muscle fatigue,[10,11] or soft-tissue damage,[12,13] due in part, to abnormal muscle recruitment patterns and atypical synchronization of motor unit firing, which are most likely influenced by the mechanical properties of a sport surface. To properly analyze the player–surface interaction, it is necessary to measure the mechanical properties of the field,[14] which was a key element of our research study. We selected ASTM F-1936 protocol to measure time-dependent properties. These particular elements are important due to their effect on muscle recruitment patterns and metabolic energy expenditure. We used International Rugby Board Regulation 22 to determine field quality where our test fields were slightly stiffer than International Rugby Board standards but within the range for vertical deformation and ER. We tested the fields in 4 different areas and consistency between sites was within Fédération Internationale de Football Association specifications.

Energy restitution was important to us because it describes the energy returned back to the athlete following an applied load, such as, the feet contacting the surface during running.[15] Energy return is a common principle used to examine player–surface interaction because returned energy from the surface allows the athlete to potentially perform a movement more efficiently. Surfaces that exhibit a relatively high energy return generally provide athletes the means to run faster and change direction more quickly.[15] Our results indicate it took 19.4 seconds longer to complete the agility protocol on the field with lower ER. The metabolic response findings show the participants expend slightly more calories on the field with greater impact absorption and where the greatest magnitude of muscle fatigue was also observed. We included a 30-m sprint test before and after the test protocol. The sprint test served 2 purposes: (1) to demonstrate the agility test course properly induced muscle fatigue and (2) to show field mechanical properties influence running speed. The posttest sprint times indicated the participants ran 9.7% slower than their pretest times, which we interpreted as muscle fatigue being a contributing factor. The pretest sprint times were 7.6% faster on the field demonstrating the greatest ER.

We must not only consider cardiopulmonary and metabolic response when examining fatigue but also muscle activation and recruitment patterns. The EMG analysis indicated the sum of muscle activity was 13% greater when performing on the field with less energy return suggesting greater muscle recruitment. The EMG analysis also revealed the distribution of upper and lower leg musculature workload was different between test fields. Lower leg

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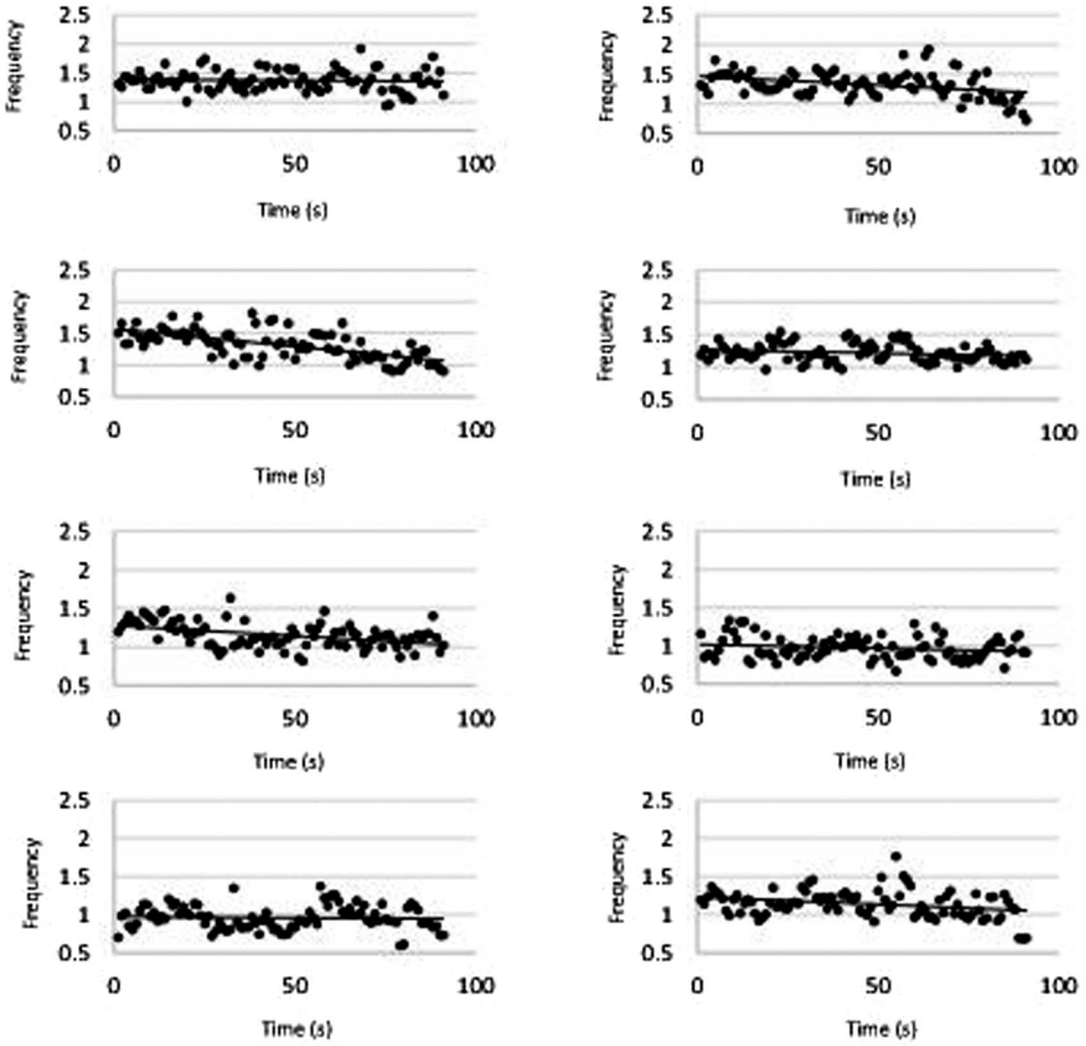

Figure 3 — Muscle frequency depicting muscle fatigue for the selected lower-extremity muscles: top to bottom, vastus medialis, biceps femoris, gastrocnemius medial head, and tibialis anterior. Field A (left column) demonstrates less energy return and greater absorption properties, whereas Field B (right column) demonstrates greater energy return and less force-attenuation properties.

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muscle contribution relative to the upper leg musculature was 80% for the stiffer field and the lower leg musculature relative contribution on the field with greater absorption properties was 64%. During a muscle fatigued state, the increased EMG amplitude and lower muscle activity on the field with greater energy return could be the result of more frequent proprioceptor activation potentially triggered by the repetitive higher loading rates.[16–19]

Our study provides evidence a sport field with a specific quantity of energy return and impact absorption causes the musculoskeletal system to produce a specific muscle recruitment pattern in order to compensate for the field's particular time-dependent properties. Identifying different adaptations of the human body influenced by the mechanical properties of a field, sport-training professionals can better prepare their athletes to meet the specific demands of a playing surface.[19,20] Based on several studies in conjunction with our findings,[7,8,21] we speculate muscle function abnormalities associated with practicing and competing on surfaces with different mechanical properties could be potential precursors for poor performance and soft-tissue-related injuries that are associated with neuromuscular fatigue.[22] The complexity of identifying factors affecting on-field player performance and injury causation should expand beyond field type (synthetic or natural) alone, ironically, the one variable not included in many player–surface analyses is *field mechanical properties*.

Practical Applications

Our findings have initiated a more comprehensive understanding of player–surface interaction and providing further knowledge of the mechanism for predisposing an individual to overuse injuries induced by muscle fatigue. Player development professionals should consider the mechanical properties of a sport surface when developing specific conditioning protocols and determining optimal training volumes for reducing the probability of early on-set muscle fatigue.[4,23,24] Preparing athletes on a surface with similar mechanical properties as the competition surface could be a solution.[21,25] Having access to a wide array of sport fields with different mechanical properties might be unreasonable for most athletic programs. However, an alternative approach for athlete preparation professionals to simulate a field with a certain range of mechanical

properties could consider the footwear–surface interaction.[16,26,27] It would be less costly for an athletic program to possess various footwear options for the athletes to elicit the desired effect on the musculoskeletal system when preparing for a competition.

Conclusions

The results of the study support our hypothesis stating the mechanical properties associated with a playing surface influences muscle activity and muscle recruitment patterns. Metabolic response demonstrated only a slight difference between fields when the athletes were most fatigued but not at a significant level, which could be associated with the duration of the test protocol. However, projecting the energy expenditure results over the entire duration of a game or match could show greater differences. In addition, the sport field that exhibited the greatest magnitude of energy return also produced the lowest sprint and agility course times. In our opinion, playing on a field with unfamiliar mechanical properties could cause an athlete to produce atypical movement strategies that might contribute to overuse of the neuromuscular system and ultimately lead to isolated muscle fatigue. In conjunction with formulating an algorithm to determine muscle activation patterns based on field mechanical properties, future research should investigate the prospect of using specialized athletic footwear to replicate specific muscle activation patterns.

Acknowledgments

The authors would like to acknowledge the support of the Office of the Vice President of Research (OVPR) at Kennesaw State University for awarding the grant that funded the research project. The findings from this investigation do not constitute endorsement by the authors or the journal of any products assessed.

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